Ralu Chineme (he/him)

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EDUCATION

University of British Columbia Expected Graduation: May 2030
Bachelor of Applied Science, Computer Engineering

- **Co-op Availability:** 4 months, May to August 2027
- **Relevant Courses:** CPEN 211 Computer Architecture (in progress: SystemVerilog on a DE10-Lite FPGA, RISC-V assembly, C), Introduction to Computation in Engineering Design (94%), Experimental Physics Labs

TECHNICAL SKILLS

Languages & Programming: C, C++, Python, SystemVerilog, RISC-V Assembly, Java, TypeScript, HTML/CSS
Hardware: DE10-Lite FPGA (Intel MAX 10), Arduino, Microcontrollers, Raspberry Pi, Sensors and Servos, PC Building & Assembly
Lab Equipment: Oscilloscope, Multimeter, DC Power Supply, Function Generator, Breadboard Prototyping
Tools & Platforms: Linux, SSH, Git/GitHub, Docker, VS Code, IntelliJ IDEA, Tailscale
Certifications: Arm Education (edX): Introduction to Microprocessors, Embedded Systems Essentials with Arm, Build Your First IoT Application on Arm, Machine Learning at the Edge; Standard First Aid + CPR-C; HCS 3000

TECHNICAL EXPERIENCE

UBC AgroBot | Vancouver, BC Sep 2026 – Present
System Architecture Subteam Member

- Preparing to build a camera plugin for the robot's NVIDIA Jetson computer by studying GStreamer multimedia pipelines through NVIDIA and official GStreamer documentation
- Contribute to System Architecture subteam meetings, coordinating with members on how the robot's subsystems are designed and connected

MFC Games | Calgary, AB May 2024 – Aug 2024
Website Administrator

- Migrated 35+ blog posts and their photos from the organization's old website to a new WordPress site, preserving the archive of past events for participants and organizers
- Coordinated updates with organizers to improve site navigation and keep content consistent across the site

TECHNICAL PROJECTS

Raspberry Pi Homelab | Personal Project Apr 2026 – Present

- Engineered a Raspberry Pi 5 homelab running six containerized services (Nextcloud, Pi-hole, Jellyfin, Uptime Kuma, Portainer, Homepage) reliably on a single device
- Configured an NVMe SSD with dedicated storage paths for cloud and media data, separating application data from the OS drive to protect against data loss
- Deployed Pi-hole as the home network DNS server and Tailscale for remote access, turning Nextcloud into a multi-user private cloud for family members
- Administer the headless Linux server over SSH, troubleshooting Docker, networking, storage, and authentication issues to keep services online

Embedded Arduino Systems | UBC Coursework Jan 2026 – Apr 2026

- Built a sensor-driven mechanical claw with a team, writing the servo rotation-control code and tuning rotation range and sensor trigger distance to prevent over-rotation and missed detections
- Programmed button-driven controllers using debouncing, edge detection, and non-blocking millis() timing, including a flashlight with 500 ms double-click detection and an emergency-blink mode
- Developed a reaction-time tester with randomized 2 to 7 second delays and a potentiometer-controlled LED chaser with adjustable speed and direction, plus a seven-segment letter display

AWARDS & INTERESTS

- **2nd Place**, MIT Battlecode High School Division (2022); also competed in Battlesnake and the JunioTech Hackathon
- **Athlete of the Year**, Master's Academy and College (2025)
- **League Champions**, Calgary Independent Schools Athletic Association (CISAA), as co-coach of the Master's Eagles Grade 7 boys' basketball team
- Assembled a custom PC for gaming and AI work, troubleshooting hardware issues during setup; exploring microprocessors and semiconductors through online courses; powerlifting